

POST PILOT.AI

Social Media Automation & Publishing Platform

Project Name	Elite Social — Social Media Automation & Publishing Platform
Project Type	Multi-Platform Social Media Scheduler & AI Content Generator
Platforms	Instagram (Meta Graph API) & LinkedIn (REST API)

Project Overview

Post Pilot.ai is a unified platform that lets content creators and digital marketers manage multiple Instagram and LinkedIn profiles from a single dashboard. Users upload media, generate AI-powered captions, schedule posts, and publish to multiple accounts simultaneously — eliminating the overhead of logging into individual platforms.

Core Objectives

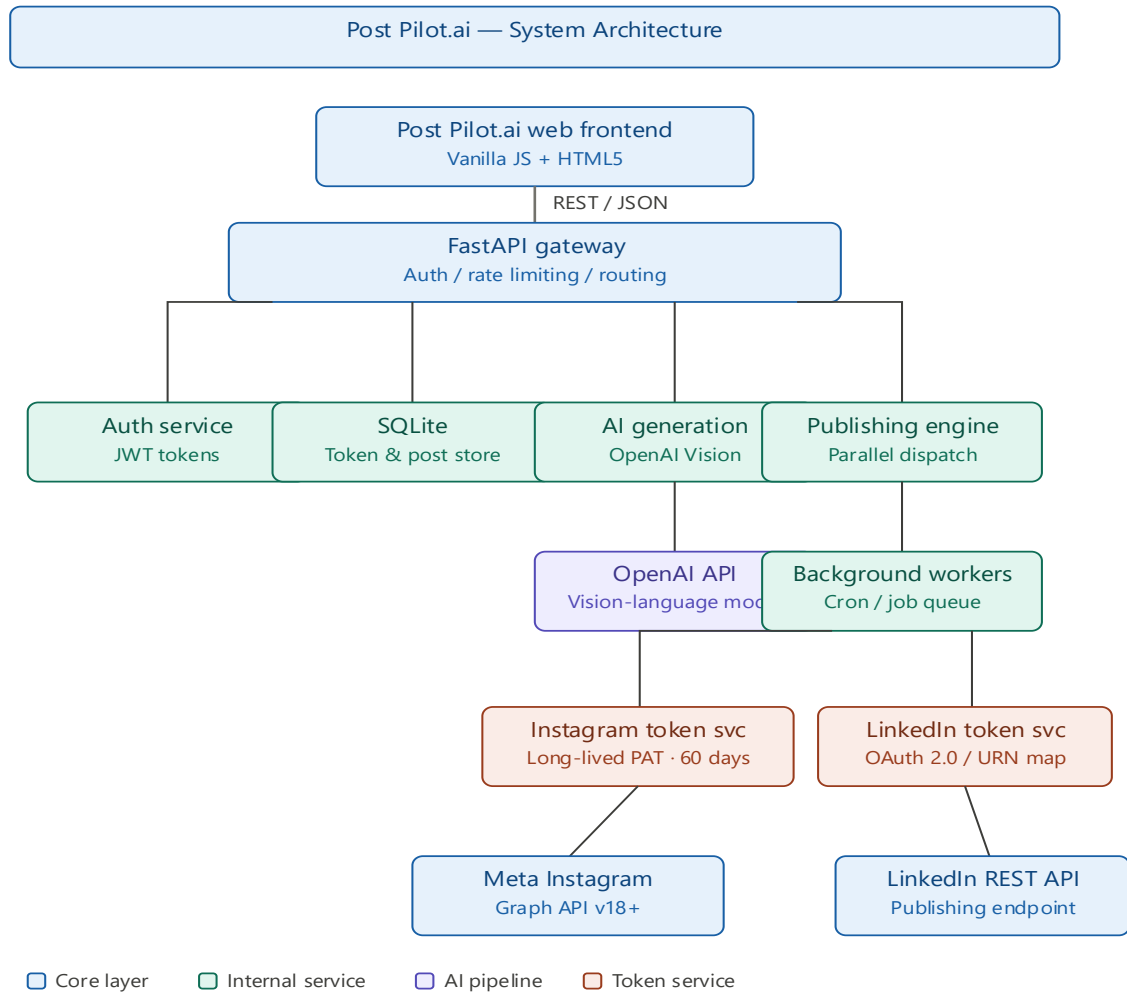
- Unified Multi-Channel Publishing:** Draft, preview, and publish to multiple Instagram and LinkedIn accounts simultaneously with one click.
- Secure OAuth 2.0 Infrastructure:** Automated Meta and LinkedIn token management with long-lived token exchange (60-day validity).
- AI Magic Engine:** Vision-language model analyses uploaded images and auto-generates captions, hashtags, and CTAs.
- Asynchronous Scheduler:** Background job queue for future-scheduled posts — runs independently without blocking the UI.
- Real-Time Diagnostics:** Live dashboard showing AI engine health, API connectivity, and storage status.

Technology Stack

Layer	Technology	Purpose
Frontend UI	Vanilla JS + HTML5 + CSS	Fast load times, micro-animations, precise styling control — no framework overhead.
Backend Core	FastAPI (Python 3.10+)	Async ASGI framework for high throughput with auto-generated OpenAPI documentation.
Database	SQLite3	Relational storage for users, posts, and encrypted tokens with ACID guarantees.
AI Vision & Gen	OpenAI	Vision-language model for image analysis and platform-specific caption generation.
Scheduler	Async Background Workers	Persistent cron loop handling scheduled post delivery without blocking the interface.
Platform APIs	Meta Graph API / LinkedIn API	OAuth 2.0 integrations with long-lived tokens and LinkedIn URN schema mapping.

System Architecture

The frontend communicates with the FastAPI Gateway via REST/JSON. The gateway delegates to four internal services: Auth, Database, AI Generation, and Publishing Engine — which dispatches posts via Instagram and LinkedIn token services, backed by background worker queues.



Core Features

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1	Unified Publisher	Upload media, write a caption, select multiple accounts, preview in native layout, and publish — all from one screen.
2	AI Magic Engine	One-click vision analysis of uploaded images. Generates platform-specific captions, hashtags, and call-to-actions via OpenAI.
3	Token Auto-Resolution	Users paste their Page token once. The backend verifies it, exchanges for a 60-day long-lived token, resolves the Instagram handle, and saves automatically.
4	Job Scheduler & Queue	Calendar time-picker for scheduling posts days in advance. Background daemon polls and publishes pending jobs via Graph API — no manual steps.
5	Real-Time Status Dashboard	Live indicators for AI engine health, media storage, and Instagram/LinkedIn API connectivity shown directly in the main dashboard.

Database Schema

Four core entities. PK = Primary Key, FK = Foreign Key.

USERS	INSTAGRAM_ACCOUNTS	LINKEDIN_ACCOUNTS	SCHEDULED_POSTS
id (PK) email full_name hashed_password created_at	instagram_account_id (PK) user_id (FK) username access_token expires_at last_refreshed_at status	linkedin_account_id (PK) user_id (FK) name access_token expires_at last_refreshed_at status	id (PK) user_id (FK) caption media_url schedule_time platforms status

Implementation Timeline

Phase	Milestone	Deliverables	Start	End	Status
P1	Infrastructure	FastAPI setup, DB migrations, environment config	18-05-26	19-05-26	Completed
P2	Frontend UI	Dark-mode mockups, Connections view, HSL stylesheets	18-05-26	31-05-26	In Progress
P3	Auth & Users	JWT endpoints, login/signup, cookie security	20-05-26	21-05-26	Completed
P4	Instagram API	Token validation, long-lived exchange, publishing	22-05-26	23-05-26	Completed
P5	LinkedIn API	URN mapping, image registration, publish pipeline	23-05-26	24-05-26	Completed
P6	AI Magic Engine	Vision model API, image analysis, caption endpoints	24-05-26	25-05-26	Completed
P7	Scheduler & Cron	Background queue daemon, calendar UI, job tracking	26-05-26	27-05-26	In Progress
P8	Analytics	Stats aggregation API, historical logs, dashboard metrics	28-05-26	29-05-26	Pending
P9	Testing & Production	Integration tests, sandbox publishes, production build locks	30-05-26	31-05-26	Pending

Conclusion

Post Pilot.ai is a modular, production-ready platform that bridges AI content generation with multi-channel social publishing. Through automated token management, vision-based caption generation, and parallel posting pipelines, it delivers a significant efficiency improvement for social media teams.